DESCRIPTION OF COURSEWORK

Course Code	EEE308
Course Name	RF&Microwave Circuits
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2023/09
Assessment Title	Microwave Calibration Technique and Band Pass Filter Measurement

A. Introduction/ Situation/ Background Information

S-parameters (scattering parameters) are used to describe the way a device modifies a signal. The FieldFox can measure all four S-parameters with a single connection. When the source comes from port 1, the measurement is said to be in the forward direction. When the source comes from port 2, the measurement is said to be in the reverse direction.

S_{11} and S_{22} reflection measurements are used to measure the amount of reflections off the corresponding DUT port. Low reflections means there is a good impedance match between the source and DUT. S_{21} and S_{12} transmission measurements are used to measure the loss or gain through a DUT over a specified frequency range. Both ends of the DUT must be connected to the FieldFox. The FieldFox signal source is transmitted out one of the test port connectors, through the DUT, and into the other test port connector.

Why and When to Calibrate

Calibration removes the systematic errors that are associated with measurements. There are well-defined and understood systematic errors that are measured and calculated during the calibration process. These errors are caused by leakage signals inside the FieldFox, by the frequency response of the FieldFox receivers, and by reflections inside the FieldFox that interact with the DUT. After calibration, these errors are removed from subsequent measurement

Definitions DUT (Device Under Test)

The cable, antenna, transmission line, amplifier, or anything else that is connected to the FieldFox that is to be measured. Calibration Standards - THRU,OPEN, SHORT, and LOAD are standards that are used during calibration. When an RF signal 'hits' these components, the signals are reflected in a predictable manner. These components can also be used to terminate a DUT port during some measurements. SHORT and OPEN standards both cause 100% of an RF signal to be reflected. The difference between these two standards is what happens to the phase of the reflected signal, which is beyond the scope of this discussion. Although an OPEN standard is a precision component, simply leaving nothing connected at the end of a cable can be a reasonable substitute for an OPEN. A LOAD standard absorbs almost ALL of the incident signal and very little signal is reflected back to the source. A THRU standard is used during some calibration steps to connect PORT 1 to PORT 2 in place of the DUT. A Flush THRU connection can be made when cables that connect with the DUT can mate with each other.. Otherwise, any reasonably short cable can be used as a THRU standard. Calibration Reference Plane is the point (or points) at which the DUT and cal standards are connected during a calibration. This can be at the FieldFox test port connectors, or at the end of jumper cables or adapters.

Assignments for this coursework: Microwave Calibration Technique and Band Pass Filter Measurement In this experiment, students shall measure and characterize the properties of the fabricated band pass filter using the N9923A FieldFox Analyzer. The mechanical calibration procedure is detailed in the FieldFox User's Guide pages 66-68.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Build microwave circuits and measure its signals

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

The report must be submitted (Moodle) before 31 December 2023. The report must include abstract ,introduction, theory, methodology, results &discussion, conclusion and references.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Lab Report– **Microwave Calibration Technique and Band Pass Filter**	10%
	TOTAL	

F. Tasks

Perform full two port mechanical calibration procedure and measure the S-parameters of a band pas filter using N9932A Field Fox Analyzer

APPENDIX 1

MARKING RUBRICS

	Excellent (90-100%)	**Good (80-89%)**	**Adequate (60-79%)**	**Poor (20-60%)**	**Lacking 20-0%**
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.

methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual.	The description gives generalities, enough for reader to understand how the experiment was conducted	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted	Part is missing or plagiarized
Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.

Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or realworld connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.